1. docker –version: Kiểm tra Docker đã cài chưa

```
Microsoft Windows [Version 10.0.19045.6216]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Student>docker --version
Docker version 24.0.6, build ed223bc
```

2. docker run hello-world: Chạy container test để xác nhận Docker hoạt động OK

```
C:\Users\Student>docker run hello-world
Unable to find image 'hello-world:latest' locally
latest: Pulling from library/hello-world
4f55086f7dd0: Pull complete
Digest: sha256:452a468a4bf985040037cb6d5392410206e47db9bf5b7278d281f94d1c2d0931
Status: Downloaded newer image for hello-world:latest

Hello from Docker!
This message shows that your installation appears to be working correctly.

To generate this message, Docker took the following steps:
 1. The Docker client contacted the Docker daemon.
 2. The Docker daemon pulled the "hello-world" image from the Docker Hub.
    (amd64)
 3. The Docker daemon created a new container from that image which runs the
    executable that produces the output you are currently reading.
 4. The Docker daemon streamed that output to the Docker client, which sent it
    to your terminal.

To try something more ambitious, you can run an Ubuntu container with:
 $ docker run -it ubuntu bash

Share images, automate workflows, and more with a free Docker ID:
 https://hub.docker.com/

For more examples and ideas, visit:
 https://docs.docker.com/get-started/


C:\Users\Student>
```

3. docker pull nginx: Tải image nginx từ Docker Hub

```
C:\Users\Student>docker pull nginx
Using default tag: latest
latest: Pulling from library/nginx
ec781dee3f47: Pull complete
bb3d0aa29654: Pull complete
510ddf6557d6: Pull complete
cde7a05ae428: Pull complete
587e3d84dbb5: Pull complete
3189680c601f: Pull complete
5e815e07e569: Pull complete
Digest: sha256:7150b3a39203cb5bee612ff4a9d18774f8c7caf6399d6e8985e97e28eb751c18
Status: Downloaded newer image for nginx:latest
docker.io/library/nginx:latest

What's Next?
  1. Sign in to your Docker account → docker login
  2. View a summary of image vulnerabilities and recommendations → docker scout quickview nginx

C:\Users\Student>
```

4. docker images: Liệt kê image trong máy

```
C:\Users\Student>docker images
REPOSITORY    TAG       IMAGE ID       CREATED        SIZE
nginx         latest    0cf1d6af5ca7   39 hours ago   161MB
hello-world   latest    e2ac70e7319a   2 days ago     10.1kB

C:\Users\Student>
```

5. docker rmi <image_id>: Xóa image

```
C:\Users\Student>docker rmi e2ac70e7319a -f
Untagged: hello-world:latest
Untagged: hello-world@sha256:452a468a4bf985040037cb6d5392410206e47db9bf5b7278d281f94d1c2d0931
Deleted: sha256:e2ac70e7319a02c5a477f5825259bd118b94e8b02c279c67afa63adab6d8685b

C:\Users\Student>
```

6. docker image prune –a: Xóa toàn bộ image không dùng

```
C:\Users\Student>docker image prune -a
WARNING! This will remove all images without at least one container associated to them.
Are you sure you want to continue? [y/N] y
Deleted Images:
untagged: nginx:latest
untagged: nginx@sha256:7150b3a39203cb5bee612ff4a9d18774f8c7caf6399d6e8985e97e28eb751c18
deleted: sha256:0cf1d6af5ca72e2ca196afdbdbe26d96f141bd3dc14d70210707cf89032ea217
deleted: sha256:61633ce66c62a864471c8b5378d9ffa3f81c766a960a3590211f3221c1ab4a59
deleted: sha256:41b747d79cc454963fb711facd1ece56107a1cea116f91f9f2226a8ad2f6aa7a
deleted: sha256:5517b4dad484d2bd81693cb01c59de8d731101babc8f4e25af94839b01cb8101
deleted: sha256:97ee1d7582029102147c004b04f96d5bca5bdd9142a91c2164f8d86069a430c3
deleted: sha256:995b22525898b5afc8b76051e02d5f4ffd7fde6aae7d2098545978c4768b73d2
deleted: sha256:d56d0c05af74d3308643ed2eb1c57c8a56ebe1922068720f393b30a285072b93
deleted: sha256:188c9b34dfbe022075d01fc4f5a305412909ef97de440783c15043e68e1b1913

Total reclaimed space: 160.9MB

C:\Users\Student>
```

7. docker run -d nginx: Chạy container nginx (chạy nền)

```
C:\Users\Student>docker run -d nginx
Unable to find image 'nginx:latest' locally
latest: Pulling from library/nginx
ec781dee3f47: Pull complete
bb3d0aa29654: Pull complete
510ddf6557d6: Pull complete
cde7a05ae428: Pull complete
587e3d84dbb5: Pull complete
3189680c601f: Pull complete
5e815e07e569: Pull complete
Digest: sha256:7150b3a39203cb5bee612ff4a9d18774f8c7caf6399d6e8985e97e28eb751c18
Status: Downloaded newer image for nginx:latest
ce7ecc1726455f43cc0a30549e2972330823b00f063f098b2dec3a6bc97a3a0a

C:\Users\Student>
```

8. docker ps: Xem container đang chạy

```
C:\Users\Student>docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED          STATUS          PORTS     NAMES
ce7ecc172645   nginx     "/docker-entrypoint.…"   27 seconds ago   Up 26 seconds   80/tcp    adoring_gagarin

C:\Users\Student>
```

9. docker ps –a: Xem tất cả container

```
C:\Users\Student>docker ps -a
CONTAINER ID   IMAGE          COMMAND                  CREATED          STATUS                     PORTS     NAMES
ce7ecc172645   nginx          "/docker-entrypoint.…"   55 seconds ago   Up 55 seconds              80/tcp    adoring_gag
arin
bf426fe7876d   e2ac70e7319a   "/hello"                 9 minutes ago    Exited (0) 9 minutes ago             zealous_sut
herland

C:\Users\Student>
```

10. docker stop <container_id>: Dừng container

```
C:\Users\Student>docker stop ce7ecc172645
ce7ecc172645

C:\Users\Student>
```

11. docker restart <container_id>: Khởi động lại container

```
C:\Users\Student>docker restart ce7ecc172645
ce7ecc172645

C:\Users\Student>
```

12. docker rm <container_id>: Xóa container

```
C:\Users\Student>docker rm ce7ecc172645 -f
ce7ecc172645

C:\Users\Student>
```

13. docker container prune: Xóa container đã dừng

```
C:\Users\Student>docker container prune
WARNING! This will remove all stopped containers.
Are you sure you want to continue? [y/N] y
Deleted Containers:
bf426fe7876d15a40a5627869189cf93914c2fb6586ee5bfcbc9bbbea9d15606

Total reclaimed space: 0B

C:\Users\Student>
```

14. docker logs <container_id>: Xem log

```
C:\Users\Student>docker logs 0949a65955fd
/docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration
/docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
/docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default.conf
10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/default.conf
/docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh
/docker-entrypoint.sh: Configuration complete; ready for start up
2026/03/26 12:56:11 [notice] 1#1: using the "epoll" event method
2026/03/26 12:56:11 [notice] 1#1: nginx/1.29.7
2026/03/26 12:56:11 [notice] 1#1: built by gcc 14.2.0 (Debian 14.2.0-19)
2026/03/26 12:56:11 [notice] 1#1: OS: Linux 5.10.102.1-microsoft-standard-WSL2
2026/03/26 12:56:11 [notice] 1#1: getrlimit(RLIMIT_NOFILE): 1048576:1048576
2026/03/26 12:56:11 [notice] 1#1: start worker processes
2026/03/26 12:56:11 [notice] 1#1: start worker process 29
2026/03/26 12:56:11 [notice] 1#1: start worker process 30
2026/03/26 12:56:11 [notice] 1#1: start worker process 31
2026/03/26 12:56:11 [notice] 1#1: start worker process 32
2026/03/26 12:56:11 [notice] 1#1: start worker process 33
2026/03/26 12:56:11 [notice] 1#1: start worker process 34
2026/03/26 12:56:11 [notice] 1#1: start worker process 35
2026/03/26 12:56:11 [notice] 1#1: start worker process 36
2026/03/26 12:56:11 [notice] 1#1: start worker process 37
2026/03/26 12:56:11 [notice] 1#1: start worker process 38
2026/03/26 12:56:11 [notice] 1#1: start worker process 39
2026/03/26 12:56:11 [notice] 1#1: start worker process 40
2026/03/26 12:56:11 [notice] 1#1: start worker process 41
```

15. docker logs -f my_nginx: Xem log realtime

16. docker exec -it <container_id> /bin/sh: Vào bên trong container

```
C:\Users\Student>docker exec -it 0949a65955fd /bin/sh
#
```

17. docker inspect <container_id>: Xem chi tiết cấu hình

```
C:\Users\Student>docker exec -it ab0454bdb951 /bin/sh
Error response from daemon: Container ab0454bdb9513e21654e0cdcfcadc6b99d58522dd16bcc03f56e08797d63a3ee is not running

C:\Users\Student>docker exec -it 0949a65955fd /bin/sh
#
C:\Users\Student>docker inspect 0949a65955fd
[
    {
        "Id": "0949a65955fd21fd695711afc19167f245aea63486763c2bb416821c4539e896",
        "Created": "2026-03-26T12:56:10.901254654Z",
        "Path": "/docker-entrypoint.sh",
        "Args": [
            "nginx",
            "-g",
            "daemon off;"
        ],
        "State": {
            "Status": "running",
            "Running": true,
            "Paused": false,
            "Restarting": false,
            "OOMKilled": false,
            "Dead": false,
            "Pid": 3592,
            "ExitCode": 0,
            "Error": "",
            "StartedAt": "2026-03-26T12:56:11.122323816Z",
            "FinishedAt": "0001-01-01T00:00:00Z"
        },
        "Image": "sha256:0cf1d6af5ca72e2ca196afdbdbe26d96f141bd3dc14d70210707cf89032ea217",
```

18. docker stats: Xem CPU/RAM

```
C:\Users\Student>docker stats
CONTAINER ID   NAME                 CPU %     MEM USAGE / LIMIT     MEM %     NET I/O       BLOCK I/O   PIDS
0949a65955fd   hardcore_northcutt   0.00%     16.48MiB / 5.943GiB   0.27%     1.16kB / 0B   0B / 0B     24
CONTAINER ID   NAME                 CPU %     MEM USAGE / LIMIT     MEM %     NET I/O       BLOCK I/O   PIDS
0949a65955fd   hardcore_northcutt   0.00%     16.48MiB / 5.943GiB   0.27%     1.16kB / 0B   0B / 0B     24
CONTAINER ID   NAME                 CPU %     MEM USAGE / LIMIT     MEM %     NET I/O       BLOCK I/O   PIDS
0949a65955fd   hardcore_northcutt   0.00%     16.48MiB / 5.943GiB   0.27%     1.16kB / 0B   0B / 0B     24
CONTAINER ID   NAME                 CPU %     MEM USAGE / LIMIT     MEM %     NET I/O       BLOCK I/O   PIDS
0949a65955fd   hardcore_northcutt   0.00%     16.48MiB / 5.943GiB   0.27%     1.16kB / 0B   0B / 0B     24
CONTAINER ID   NAME                 CPU %     MEM USAGE / LIMIT     MEM %     NET I/O       BLOCK I/O   PIDS
0949a65955fd   hardcore_northcutt   0.00%     16.48MiB / 5.943GiB   0.27%     1.16kB / 0B   0B / 0B     24
CONTAINER ID   NAME                 CPU %     MEM USAGE / LIMIT     MEM %     NET I/O       BLOCK I/O   PIDS
0949a65955fd   hardcore_northcutt   0.00%     16.48MiB / 5.943GiB   0.27%     1.16kB / 0B   0B / 0B     24
CONTAINER ID   NAME                 CPU %     MEM USAGE / LIMIT     MEM %     NET I/O       BLOCK I/O   PIDS
0949a65955fd   hardcore_northcutt   0.00%     16.48MiB / 5.943GiB   0.27%     1.16kB / 0B   0B / 0B     24
CONTAINER ID   NAME                 CPU %     MEM USAGE / LIMIT     MEM %     NET I/O       BLOCK I/O   PIDS
0949a65955fd   hardcore_northcutt   0.00%     16.48MiB / 5.943GiB   0.27%     1.16kB / 0B   0B / 0B     24
CONTAINER ID   NAME                 CPU %     MEM USAGE / LIMIT     MEM %     NET I/O       BLOCK I/O   PIDS
0949a65955fd   hardcore_northcutt   0.00%     16.48MiB / 5.943GiB   0.27%     1.16kB / 0B   0B / 0B     24
CONTAINER ID   NAME                 CPU %     MEM USAGE / LIMIT     MEM %     NET I/O       BLOCK I/O   PIDS
0949a65955fd   hardcore_northcutt   0.00%     16.48MiB / 5.943GiB   0.27%     1.16kB / 0B   0B / 0B     24
CONTAINER ID   NAME                 CPU %     MEM USAGE / LIMIT     MEM %     NET I/O       BLOCK I/O   PIDS
0949a65955fd   hardcore_northcutt   0.00%     16.48MiB / 5.943GiB   0.27%     1.16kB / 0B   0B / 0B     24
CONTAINER ID   NAME                 CPU %     MEM USAGE / LIMIT     MEM %     NET I/O       BLOCK I/O   PIDS
0949a65955fd   hardcore_northcutt   0.00%     16.48MiB / 5.943GiB   0.27%     1.16kB / 0B   0B / 0B     24
CONTAINER ID   NAME                 CPU %     MEM USAGE / LIMIT     MEM %     NET I/O       BLOCK I/O   PIDS
0949a65955fd   hardcore_northcutt   0.00%     16.48MiB / 5.943GiB   0.27%     1.16kB / 0B   0B / 0B     24
CONTAINER ID   NAME                 CPU %     MEM USAGE / LIMIT     MEM %     NET I/O       BLOCK I/O   PIDS
0949a65955fd   hardcore_northcutt   0.00%     16.48MiB / 5.943GiB   0.27%     1.16kB / 0B   0B / 0B     24
CONTAINER ID   NAME                 CPU %     MEM USAGE / LIMIT     MEM %     NET I/O       BLOCK I/O   PIDS
0949a65955fd   hardcore_northcutt   0.00%     16.48MiB / 5.943GiB   0.27%     1.16kB / 0B   0B / 0B     24
CONTAINER ID   NAME                 CPU %     MEM USAGE / LIMIT     MEM %     NET I/O       BLOCK I/O   PIDS
```

19. docker run -d -p 8080:80 nginx: Map port (host:container)

```
C:\Users\Student>docker run -d -p 8080:80 nginx
3c38dc81c7ca623b26795b57fd3dfb5d5f5f59abd92d3790fff811fdf999df30

C:\Users\Student>
```

20. docker run -d --name my_nginx nginx: Đặt tên container

```
C:\Users\Student>docker run -d --name my_nginx nginx
fbd3d450cc8c29ebd0ea46eafe114023799a95256a1ce7398260c88207d76868

C:\Users\Student>
```

21. docker run -d -v mydata:/data nginx: Tạo volume để lưu dữ liệu

```
C:\Users\Student>docker run -d -v mydata:/data nginx
06f4d3ab90b7277be5cceb63c076e23b9fdb1f3a5b23516c07ca8c7adefaa770

C:\Users\Student>
```

22. docker volume ls: Xem danh sách

```
C:\Users\Student>docker volume ls
DRIVER    VOLUME NAME
local     mydata

C:\Users\Student>
```

22. docker volume prune: Xóa volume không dùng

```
C:\Users\Student>docker volume prune
WARNING! This will remove anonymous local volumes not used by at least one container.
Are you sure you want to continue? [y/N] y
Total reclaimed space: 0B

C:\Users\Student>
```

23. docker network ls: Xem network

```
C:\Users\Student>docker network ls
NETWORK ID     NAME      DRIVER    SCOPE
09c5bbfa3124   bridge    bridge    local
3ca57c3e562c   host      host      local
4ef757842cb0   none      null      local

C:\Users\Student>
```

24. docker network create my_network: Tạo network riêng

```
C:\Users\Student>docker network ls
NETWORK ID     NAME      DRIVER    SCOPE
09c5bbfa3124   bridge    bridge    local
3ca57c3e562c   host      host      local
4ef757842cb0   none      null      local

C:\Users\Student>docker network create my_network
a9983c4bc7e72f7395f318a13774b88eee4fc6f2527a7ba927714a6dfb64de18

C:\Users\Student>
```

25. docker run -d --network my_network --name my_container nginx: Chạy container trong network

```
C:\Users\Student>docker run -d --network my_network --name my_container nginx
b231dc32124b09b12c64b05a7350d5db0f69369d12740cc25b32d6f22c8fc815

C:\Users\Student>
```

26. docker network connect my_network my_nginx: Kết nối container vào network

```
C:\Users\Student>docker network connect my_network my_nginx

C:\Users\Student>
```

27. docker run -d -e MY_ENV=hello_world nginx: Truyền biến môi trường vào container

```
C:\Users\Student>docker run -d -e MY_ENV=hello_world nginx
ae1e94767b8ff4d45d8dce18d8f8aec129550c787494833dce442a806b56b6b2

C:\Users\Student>
```

28.

FROM nginx: Dùng nginx làm base image

COPY index.html /usr/share/nginx/html/index.html: Copy file vào container

29. docker build -t my_nginx_image: Build image mới

```
C:\Users\Student\Desktop>rename Dockerfile.txt Dockerfile

C:\Users\Student\Desktop>docker build -t my_nginx_image .
[+] Building 0.2s (7/7) FINISHED                                                                    docker:default
 => [internal] load build definition from Dockerfile                                                          0.0s
 => => transferring dockerfile: 97B                                                                           0.0s
 => [internal] load .dockerignore                                                                             0.0s
 => => transferring context: 2B                                                                               0.0s
 => [internal] load metadata for docker.io/library/nginx:latest                                               0.0s
 => [internal] load build context                                                                             0.0s
 => => transferring context: 64B                                                                              0.0s
 => [1/2] FROM docker.io/library/nginx                                                                        0.1s
 => [2/2] COPY index.html /usr/share/nginx/html/index.html                                                    0.0s
 => exporting to image                                                                                        0.0s
 => => exporting layers                                                                                       0.0s
 => => writing image sha256:3d6173d28c3dd6c282f0b3720ea698049afcadf0bc7eb51cb25d4f68a2243c04                  0.0s
 => => naming to docker.io/library/my_nginx_image                                                             0.0s

What's Next?
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  2. View a summary of image vulnerabilities and recommendations → docker scout quickview

C:\Users\Student\Desktop>
```

30. docker run -d -p 8080:80 my_nginx_image: Chạy image tự build

```
C:\Users\Student\Desktop>docker run -d -p 8080:80 my_nginx_image
8190bce7a4a1bbe75419b632bb39b0eba54f65598318641fdcde0d700edfccd1

C:\Users\Student\Desktop>
```